Multifunction Power Supply



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Many embedded systems these days need +5V power supply with some special functions such as power-fail detection, zero-crossing signals for mains power supply and possibility to maintain the charging of built-in batteries.

Presented here is the circuit of such a power supply for embedded systems. It provides +5V, battery charger, zero-crossing signals and power-fail signal. The circuit is based on popular low-cost components. It needs simple adjustment with potmeter POT1 to immediately get you started.

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	10V-18V (approx.)
TP3	100 Hz square wave
TP4	100 Hz square wave
TP5	100 Hz square wave
TP6	5V when power fails

Circuit and working

The circuit shown in Fig. 1 is built around a step-down transformer (X1), two bridge rectifiers (BR1 and BR2), adjustable voltage regulator LM317 (IC1), 5V voltage regulator 7805 (IC2), hex inverter Schmitt trigger 74HC14 (IC3) and a few discrete components.

The mains power supply 230V AC, 50Hz is applied to connector CON1. Fuse F1 protects the input from overloads. Resistor R1 and capacitor C1 filter some of the noise coming from the mains power supply. The secondary of power transformer X1 is 12V, 2.5A.

CON3 provides +5V, 1A power supply using standard 5V regulator IC2. Rectifying bridge BR1 is of 2.5A. The value of the main filtering capacitor (C4) should be at least 4700 µF. IC2 provides regulated +5V with output current up to 1A. If a higher current is needed, regulators like 78T05 (3A, 5V) or 78S05 (2A, 5V) can be used. In practice, it is better to limit the load for 78XX up to around 0.7-0.8A. The unregulated output at CON4 provides voltage of 10V to 18V depending on the transformer used and the current consumption from the power supply. This unregulated voltage can be used for peripheral functions. Fuse F2 is used to protect the output.

The power supply incorporates battery charger with adjustable regulator IC1. A 6V rechargeable battery is used to provide power supply to some parts of the system when the mains power supply does not function properly. The maximum voltage level across the rechargeable battery is adjusted using potmeter POT1. The maximum charging current is limited by resistor R7. Regulator IC1 can provide output voltage of +1.25V to +8.2V, adjustable with potmeter POT1.

Diodes D1, D2 and D3 protect regulators IC1 and IC2. Bridge rectifier BR2 is used only to provide signals around the zero crossings. Capacitor C2 should have a small value.

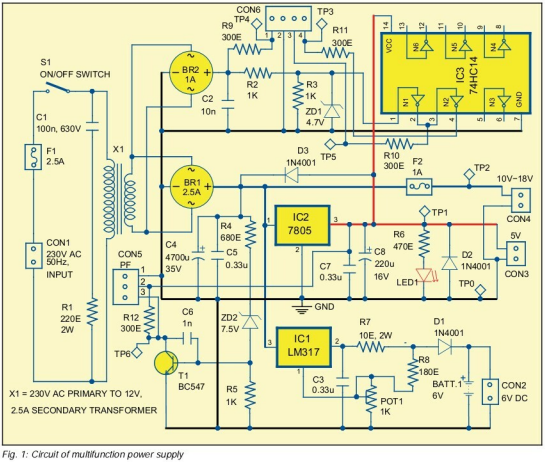


Fig. 1: Circuit of multifunction power supply